4. Linear transformations and diagonalization

This chapter is devoted to analyzing linear transformations between abstract vector spaces. The "high point" of this analysis is the so called Rank-Nullity-Theorem in its various forms. We will then focus on the case of a linear operator (that is, the case where the source and target vector spaces coincide). Here, we will see how diagonalization can help us to understand the "action" of a linear operator.

4.1. Linear transformations

We know the definition of a linear transformation $T\colon \mathbb{R}^n \to \mathbb{R}^m$.They are mappings that preserve sums and scalar multiples. More precisely, $T(\mathbf{x}+\mathbf{y}) = T(\mathbf{x})+T(\mathbf{y})$ and $T(c\mathbf{x}) = cT(\mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $c \in \mathbb{R}$. This definitions obviously makes sense for arbitrary vector spaces:

4.1 Definition. Let V, W be $\mathbb{F}$-vector spaces (that is, V, W are *both* either real or complex vector spaces). A *linear transformation* (also: linear map) from V to W is a map

$$T\colon V \to W$$

such that

L1 $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$ for all $\mathbf{v}, \mathbf{w} \in V$.

L2 $T(c\mathbf{v}) = cT(\mathbf{v})$ for all $\mathbf{v} \in V$ and $c \in \mathbb{F}$.

If $V = W$, then such a $T\colon V \to V$ is also called a *linear operator*.

4.2 Examples.

1. As before if $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$, and A is any real $m \times n$-matrix then $T_A\colon \mathbb{R}^n \to \mathbb{R}^m$ given by $T_A(\mathbf{x}) = A\mathbf{x}$ is a linear transformation. Recall that all linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$ are of this form.

2. The first example generalizes immediately to linear transformations $\mathbb{C}^n \to \mathbb{C}^m$: if A is an $m \times n$-matrix with complex entries then $T_A\colon \mathbb{C}^n \to \mathbb{C}^m$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$ (with matrix multiplication defined in the same way as for real matrices) is a linear transformation. Again, all linear transformations from $\mathbb{C}^n$ to $\mathbb{C}^m$ are of this form.

3. Let $V = \mathcal{C}^1(I)$ be the space of functions that are differentiable on the interval I such that the derivative is continuous; let $W = \mathcal{C}(I)$ the space of continuous functions on I. Let $D\colon V \to W$ be defined as
$$D(f) = f' = \frac{df}{dx}.$$
Then D is a linear transformation.

   Indeed, calculus tells us that if $f, g \in V$, then $f+g$ is differentiable and $(f+g)'(x) = f'(x)+g'(x)$ for all $x \in I$. Similarly, for any constant c, cf is differentiable and $(cf)'(x) = c \cdot f'(x)$ for all $x \in I$.

   Note if $I = [a, b]$, there is also a well-known linear transformation $\mathrm{Int}\colon W \to V$ defined by
$$\mathrm{Int}_a(f)(x) = \int_a^x f(s)ds.$$
(The Fundamental Theorem of Calculus guarantees that $\mathrm{Int}_a(f)$ is indeed in $\mathcal{C}^1(I)$.)

4. Let V be a real vector space with basis $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$. Then the *coordinate mapping* $V \to \mathbb{R}^n$ associating to $\mathbf{v}$ its coordinate vector $[\mathbf{v}]_{\mathcal{B}}$ is a linear transformation. (cf. parts 1 and 2 in Facts 3.14

5. — Let V be a complex vector space and $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in V$. Then $T\colon \mathbb{C}^k \to V$ defined by
$$T\left(\begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_k \end{bmatrix}\right) = z_1\mathbf{v}_1 + z_2\mathbf{v}_2 + \cdots + z_k\mathbf{v}_k$$
is a linear transformation.

6. Let $V = \mathcal{C}(\mathbb{R})$ and $p \in \mathbb{R}$ arbitrary. Then
$$\mathrm{ev}_p\colon V \to \mathbb{R}$$
defined by $\mathrm{ev}_p(f(x)) = f(p)$ is a linear transformation.

7. If V, W are arbitrary $\mathbb{F}$-vector spaces there is always the zero-transformation $V \to W$ defined by $\mathbf{v} \mapsto \mathbf{0}$.

8. For any scalar $r \in \mathbb{F}$, the map $V \to V$ that sends $\mathbf{v}$ to $r\mathbf{v}$ is a linear transformation.

9. Suppose $F\colon \mathbb{R}^n \to \mathbb{R}^m$ is a funtion and $p \in \mathbb{R}^n$. Then F is called differentiable at p, if there is a linear transformation $T\colon \mathbb{R}^n \to \mathbb{R}^m$ such that
$$F(p + \mathbf{x}) = F(p) + T(\mathbf{x}) + o(\mathbf{x})$$

and $o(\mathbf{x})$ is "small" in the sense that

$$\lim_{\mathbf{x}\to\mathbf{0}} \frac{|o(\mathbf{x})|}{|\mathbf{x}|} = 0.$$

Here for any $\mathbf{y} \in \mathbb{R}^p$, $|\mathbf{y}|$ is the *length* of $\mathbf{y}$ defined as $\sqrt{\sum_{i=1}^p y_i^2}$. What this means is that there exists a linear transformation T such that the "error term" $F(p+\mathbf{x})-F(p)-T(\mathbf{x}) = o(\mathbf{x})$ is very small as long as $\mathbf{x}$ has small entries. In case $n = m = 1$ this is the usual definition of differentiability.

The point is, that for small $\mathbf{x}$, the difference $F(p+\mathbf{x}) - F(p)$ can be approximated well by $T(\mathbf{x})$.

(In the case that $n = m = 1$, this becomes the usual definition of differentiability at a point $p \in \mathbb{R}$: indeed, in this case any linear transformation $T\colon \mathbb{R} \to \mathbb{R}$ is T_A for A some 1×1-matrix, so it is of the form $x \mapsto cx$ for a unique constant c (depending on T). The above definition then says that a function $f\colon \mathbb{R} \to \mathbb{R}$ is differentiable at $p \in \mathbb{R}$ if and only if there is $c \in \mathbb{R}$ such that

$$f(p+x) = f(p) + cx + o(x)$$

and

$$\lim_{x\to 0} \frac{|o(x)|}{|x|} = 0.$$

Solving for c, this means that

$$c = \lim_{x\to 0} \frac{f(p+x) - f(x)}{x}$$

which you may have seen as the definition of $f'(p)$. More geometrically, it simply says that the graph of the linear function $f(p) + cx$ is a tangent to the graph of f at the point $(p, f(p))$.)

10. As mentioned before, in Quantum Mechanics, the possible states of a system are certain vectors in a complex vector space H (which depends on the system in question).

    In this model, "observables," that is quantities that can be measured in experiments, correspond to certain linear operators $T\colon H \to H$. The story is roughly as follows (depending on the exact interpretation of QM you adopt): assuming H is finite dimensional (which is true only for very simple systems), the *eigenvectors*[1] of H are the states that the system is in *after the measurement*, and the corresponding *eigenvalue* is the result of the measurement. As an example, the possible states the *spinA quantum mechanical*

[1]We have not yet defined the notion of an eigenvector or eigenvalue of a linear operator on an arbitrary vector space H; however it is what you would expect: an eigenvector is a vector $\mathbf{v} \neq \mathbf{0} \in H$ such that $T(\mathbf{v}) = \lambda\mathbf{v}$ for some $\lambda \in \mathbb{C}$, and λ is called an eigenvalue of T.

analogue of magnetic momentum. of an electron can have are described by a two dimensional complex vector space, which we think of as $H = \mathbb{C}^2$. The vector $\mathbf{e}_1$ should signify the state "spin up" and the vector $\mathbf{e}_2$ should signify the state "spin down" (relative to some fixed given spacial axis). Associated to "measuring spin along this axis" is then the linear transformation T_A with

$$A = \begin{bmatrix} \frac{1}{2} & \\ & \frac{-1}{2} \end{bmatrix}$$

So $\mathbf{e}_1$ and $\mathbf{e}_2$ are eigenvectors with eigenvalues $\pm\frac{1}{2}$, respectively. If spin is measured, and the outcome is $\frac{1}{2}$, then we know that the system is in state $\mathbf{e}_1$ (spin up), and the "amount" of spin it has is $\frac{1}{2}$ (in appropriate units). Similarly, a measurement of $\frac{-1}{2}$ means the system is in state $\mathbf{e}_2$ (spin down), and the amount is $\frac{-1}{2}$ (oriented versus the axis). No other outcomes are possible, since T_A has only two eigenvalues.

Note that as in the case of matrix transformation, the set of vectors $\mathbf{v} \in V$ for which $T(\mathbf{v}) = \mathbf{0}$ plays an important role for the linear transformation $T\colon V \to W$.

4.3 Proposition-Definition. *Let $T\colon V \to W$ be a linear transformation. Then its* null space *or* kernel*is defined as*

$$\mathrm{Nul}(T) = \ker T = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}\}.$$

It is a subspace of V.

Proof.

1. $\mathbf{0} \in \mathrm{Nul}(T)$: Indeed, $T(\mathbf{0}) = T(\mathbf{0}+\mathbf{0}) = T(\mathbf{0}) + T(\mathbf{0})$. By cancelation, $T(\mathbf{0}) = \mathbf{0}$.
2. If $\mathbf{v}, \mathbf{w} \in \mathrm{Nul}(T)$, then also $\mathbf{v}+\mathbf{w} \in \mathrm{Nul}(T)$: $T(\mathbf{v}+\mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w}) = \mathbf{0}+\mathbf{0} = \mathbf{0}$.
3. If $\mathbf{v} \in \mathrm{Nul}(T)$ and $c \in \mathbb{F}$, then $c\mathbf{v} \in \mathrm{Nul}(T)$: $T(c\mathbf{v}) = cT(\mathbf{v}) = c\mathbf{0} = \mathbf{0}$.

□

4.4 Example. Let V, W, D be as in Example 4.2 3 Then $\mathrm{Nul}(D) = \{f \mid D(f) = \mathbf{0}\} = \{f \mid f \text{ is constant}\}$. In particular, $\dim \mathrm{Nul}(D) = 1$ because any $f \in \mathrm{Nul}(D)$ is a scalar multiple of the constant function $1\colon I \to \mathbb{R}$ that sends everything to 1.

Recall that the null space of a matrix A gave us important information: if $\mathrm{Nul}(A) = \{\mathbf{0}\}$, then the columns of A are linearly independent. Equivalently, every system

$$A\mathbf{x} = \mathbf{b}$$

has *at most one* solution. This is the same as saying the corresponding matrix transformation T_A is *injective* (one-to-one, but not necessarily onto): If $T_A(\mathbf{x}) = T_A(\mathbf{y})$, then $\mathbf{x} = \mathbf{y}$. Equivalently, if $\mathbf{x} \neq \mathbf{y}$, then also $T_A(\mathbf{x}) \neq T_A(\mathbf{y})$. (Indeed, if $A\mathbf{x} = A\mathbf{y}$, then $A(\mathbf{x}-\mathbf{y}) = \mathbf{0}$, so $\mathbf{x}-\mathbf{y}$ is in the null space of A. If that one is equal to $\{\mathbf{0}\}$, this says $\mathbf{x}-\mathbf{y} = \mathbf{0}$, or $\mathbf{x} = \mathbf{y}$.)

The general statement is completely analogous.

4.5 Proposition. *A linear transformation* $T: V \to W$ *is* injective *if and only if* $\mathrm{Nul}(T) = \{\mathbf{0}\}$.

Proof. This is very similar to the case of matrix transformations. If T is injective, then $T(\mathbf{x}) = T(\mathbf{y})$ iff $\mathbf{x} = \mathbf{y}$. Hence if $T(\mathbf{x}) = \mathbf{0}_W$, then $\mathbf{x} = \mathbf{0}_V$, since $T(\mathbf{0}_V) = \mathbf{0}_W$.

For the converse, if $\mathrm{Nul}(T)$ is $\{\mathbf{0}_V\}$, then if $T(\mathbf{x}) = T(\mathbf{y})$ it follows that

$$\mathbf{0} = T(\mathbf{x}) + (-T(\mathbf{y})) = T(\mathbf{x}) + (-1)T(\mathbf{y}) = T(\mathbf{x}) + T((-1)\mathbf{y}) = T(\mathbf{x} + (-1)\mathbf{y}) = T(\mathbf{x} - \mathbf{y}).$$

Hence $\mathbf{x} - \mathbf{y} \in \mathrm{Nul}(T) = \{\mathbf{0}\}$ which means $\mathbf{x} = \mathbf{y}$. Thus T is injective. □

Example. Note that D above is not injective. Indeed, $D(f) = D(f + C)$ where C is any constant function. On the other hand, Int_a is injective: $\mathrm{Int}_a(f) = \mathbf{0}$ only if $f = \mathbf{0}$: indeed, this follows immediately from $D(\mathrm{Int}(f)) = f$ (which is part of the Fundamental Theorem of Calculus):

If $\mathrm{Int}_a(f) = \mathbf{0}$, then $\mathrm{Int}_a(f)(x) = 0$ for all $x \in \mathbb{R}$. Hence its derivative $D(\mathrm{Int}_a(f))$ is the constant zero function. So $f = D(\mathrm{Int}_a(f)) = \mathbf{0}$.

There is another important subspace associated to T: the range or *image* of T, denoted $\mathrm{im}(T)$, is defined as the set of all $\mathbf{w} \in W$ for which there is $\mathbf{v}$ such that $T(\mathbf{v}) = \mathbf{w}$. Hence

$$\mathrm{im}(T) = \{T(\mathbf{v}) \mid \mathbf{v} \in V\}.$$

Where $\ker T$ is the analogue of $\mathrm{Nul}(A)$ for a matrix A, $\mathrm{im}(T)$ is similar to $\mathrm{Col}(A)$:

4.6 Lemma. $\mathrm{im}(T)$ *is a subspace of* W. *Moreover, if* V *is spanned by* $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$, *then* $\mathrm{im}(T)$ *is spanned by* $T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_p)$.

This is a homework problem.

4.7 Fact. Let U, V, W be $\mathbb{F}$-vector spaces and let $S: U \to V$ and $T: V \to W$ be linear transformations. Then $T \circ S: U \to W$ is also a linear transformation.

Proof. Exercise. □

Let us return to Example 4.2 5: Note if A is a matrix with entries in $\mathbb{F}$ and k columns we defined $A\mathbf{x}$ as the linear combination of the columns of A with the entries of $\mathbf{x}$ as coefficients. Something similar we can do in general: let V be an $\mathbb{F}$-vector space and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ be some elements in V giving an ordered list $L = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$. Following M. Artin's book "Algebra", for $\mathbf{x} \in \mathbb{F}^k$ we formally write

$$L\mathbf{x} := x_1\mathbf{v}_2 + x_2\mathbf{v}_2 + \dots x_k\mathbf{v}_k.$$

Note this notation is very conventient: if $\mathcal{B}$ is basis of V, then

$$\mathcal{B}[\mathbf{v}]_{\mathcal{B}} = \mathbf{v}.$$

Example 4.2 5 asserts that $T: \mathbb{F}^k \to V$ defined by $T(\mathbf{x}) = L\mathbf{x}$ is a linear transformation.

4.8 Example (Important Example: change of coordinates)**.** Let V be an $\mathbb{F}$ vector space of dimension n, and let $\mathcal{B}, \mathcal{C}$ be two bases of V. We have a linear transformation $S\colon \mathbb{F}^n \to V$ given by $S(\mathbf{x}) = \mathcal{B}\mathbf{x}$. Likewise we have a linear transformation $T\colon V \to \mathbb{F}^n$ defined by $T(\mathbf{v}) = [\mathbf{v}]_{\mathbb{C}}$. By Fact 4.7, the composition $\tau = T \circ S\colon \mathbb{F}^n \to \mathbb{F}^n$ is a linear transformation. How should we interpret τ?

$$\tau(\mathbf{x}) = [\mathcal{B}\mathbf{x}]_{\mathcal{C}}.$$

Thus, $\tau(\mathbf{x})$ gives us the coordinate vector of the vector $\mathbf{v} = \mathcal{B}\mathbf{x}$ with respect to the "new" basis $\mathcal{C}$ as a function of the "old" coordinates $\mathbf{x} = [\mathbf{v}]_{\mathcal{B}}$.

Note, that τ is a matrix transformation. Its standard matrix $P := [\tau]$ is called the *change of basis matrix* for the change from $\mathcal{B}$ to $\mathcal{C}$. It has the property

$$[\mathbf{v}]_{\mathcal{C}} = P[\mathbf{v}]_{\mathcal{B}}$$

for all $\mathbf{v} \in V$. We will return to this situation in the next section.

4.2. Linear transformation and coordinates

4.2.1. The matrix of a transformation

In the following we will study how coordinates (that is, the choice of bases) helps us in analyzing a linear transformation.

4.9 Example. Let $V = \mathcal{P}(\mathbb{R})_5$, $W = \mathcal{P}(\mathbb{R})_4$ and $D\colon V \to W$ be defined as

$$D(f) = f'.$$

Let $\mathcal{B} = (1, x, x^2, x^3, x^4, x^5)$ is the standard basis for V, and $\mathcal{C} = (1, x, x^2, x^3, x^4)$ the standard basis of W.

The question we're asking is, can we somehow compute the coordinate vector $[D(f)]_{\mathcal{C}}$, if we know the coordinate vector $[f]_{\mathcal{B}}$?

Note that if $f = c_0 + c_1 x + c_2 x^2 + \cdots + c_5 x^5$, then $D(f) = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + 5c_5 x^4$. Hence

$$[D(f)]_{\mathcal{C}} = \begin{bmatrix} c_1 \\ 2c_2 \\ 3c_3 \\ 4c_4 \\ 5c_5 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 3 \\ 0 \\ 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 4 \\ 0 \end{bmatrix} + c_5 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 5 \end{bmatrix}.$$

This is the same as

$$\begin{bmatrix} 0 & 1 & & & & \\ & 0 & 2 & & & \\ & & 0 & 3 & & \\ & & & 0 & 4 & \\ & & & & 0 & 5 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix}$$

So with

$$A = \begin{bmatrix} 0 & 1 & & & & \\ & 0 & 2 & & & \\ & & 0 & 3 & & \\ & & & 0 & 4 & \\ & & & & 0 & 5 \end{bmatrix}$$

we find that

$$[D(f)]_{\mathcal{C}} = A[f]_{\mathcal{B}}.$$

This is no accident. Indeed, $D(x^i) = ix^{i-1}$ for $i = 1, 2, 3, 4, 5$ and $D(1) = 0$. Hence

$$[D(x^i)]_{\mathcal{C}} = i\mathbf{e}_{i-1}$$

if $i > 0$ and $[D(1)]_{\mathcal{C}} = \mathbf{0}$, where $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4, \mathbf{e}_5$ denote the elements of the standard basis for $\mathbb{R}^5$ (in order).

Now if $f = c_0 \cdot 1 + c_1 x + \cdots + c_5 x^5$, and because D is a linear transformation and therefore maps linear combinations to linear combinations,

$$\begin{aligned} [D(f)]_{\mathcal{C}} &= c_0[D(1)]_{\mathcal{C}} + c_1[D(x)]_{\mathcal{C}} + \cdots + c_5[D(x^5)]_{\mathcal{C}} \\ &= \begin{bmatrix} [D(1)]_{\mathcal{C}} & [D(x)]_{\mathcal{C}} & \ldots & [D(x^5)]_{\mathcal{C}} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix} = A[f]_{\mathcal{B}}. \end{aligned}$$

So for instance if $f = 2 + 3x - x^2 + x^3 - x^4 + 7x^5$, then to compute $D(f) = f'$ we could proceed as follows: first determine $[f]_{\mathcal{B}}$:

$$[f]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 1 \\ -1 \\ 7 \end{bmatrix}$$

Then compute

$$[D(f)]_{\mathcal{B}} = A[f]_{\mathcal{B}} = \begin{bmatrix} 3 \\ -2 \\ 3 \\ -4 \\ 35 \end{bmatrix}$$

and finally use $[D(f)]_{\mathcal{C}}$ to get $D(f)$:

$$D(f) = 3 \cdot 1 + (-2) \cdot x + 3 \cdot x^2 - 4 \cdot x^3 + 35 \cdot x^4,$$

which in this case of course can be obtained also directly.

We can now easily determine a basis for $\mathrm{Nul}(D)$:

$$\mathbf{v} \in \mathrm{Nul}(D) \text{ if and only if } [\mathbf{v}]_{\mathcal{B}} \in \mathrm{Nul}(A).$$

Clearly $(\mathbf{e}_1)$ is a basis for $\mathrm{Nul}(A)$ and thus the constant function 1 forms a basis for $\mathrm{Nul}(D)$.

To find a basis for $\mathrm{im}(D)$, note that $\mathrm{Col}(A) = \mathbb{R}^5$ (we have 5 pivots, so $\dim \mathrm{Col}(A) = 5$ and hence $\mathrm{Col}(A) = \mathbb{R}^5$, the only subspace of $\mathbb{R}^5$ of dimension 5).

Now

$$\mathbf{w} \in \mathrm{im}(D) \text{ if and only if } [\mathbf{w}]_{\mathcal{C}} \in \mathrm{Col}(A).$$

Hence $\mathcal{C}$ is a basis for $\mathrm{im}(D) = W$.

Nothing in this example was really special to the situation at hand. So let us consider the general case. Let $T\colon V \to W$ be a linear transformation, where V, W are finite-dimensional vector spaces. Let $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ be a basis for V and let $\mathcal{C} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m)$ be a basis for W.

We may ask the following question: can we compute the coordinate vector $[T(\mathbf{v})]_{\mathcal{C}}$ if we know $[\mathbf{v}]_{\mathcal{B}}$? The answer is, in principle, yes: in the same way, we computed the standard matrix of a linear transformation $\mathbb{R}^n \to \mathbb{R}^m$, we can compute a matrix for T:

4.10 Definition. The *matrix of T* with respect to the two bases $\mathcal{B}$ and $\mathcal{C}$, denoted ${}_{\mathcal{C}}[T]_{\mathcal{B}}$, is defined as the matrix with the n columns formed by the coordinate vectors $[T(v_i)]_{\mathcal{C}}$. Thus,

$${}_{\mathcal{C}}[T]_{\mathcal{B}} = \begin{bmatrix} [T(\mathbf{v}_1)]_{\mathcal{C}} & [T(\mathbf{v}_2)]_{\mathcal{C}} & \dots & [T(\mathbf{v}_n)]_{\mathcal{C}} \end{bmatrix}.$$

It is an $m \times n$-matrix with entries in $\mathbb{F}$.

> The columns of ${}_{\mathcal{C}}[T]_{\mathcal{B}}$ are given by the coordinate vectors of the images of the basis vectors.

What is this matrix good for?

4.11 Fact. For any $\mathbf{v} \in V$, we have

$$[T(\mathbf{v})]_{\mathcal{C}} = {}_{\mathcal{C}}[T]_{\mathcal{B}} \cdot [\mathbf{v}]_{\mathcal{B}} \tag{4.1}$$

In other words, if $\mathbf{y} = [T(\mathbf{v})]_{\mathcal{C}}$ and $\mathbf{x} = [\mathbf{v}]_{\mathcal{B}}$, $A = {}_{\mathcal{C}}[T]_{\mathcal{B}}$, then

$$A\mathbf{x} = \mathbf{y}.$$

In that sense, as soon as we chose bases in V and W, T "looks" like a matrix transformation. All questions we might have can be answered by answering the same question about a matrix transformation.

IX Problem. Verify that (4.1) uniquely determines the matrix, that is, if a matrix A satisfies that for all $\mathbf{v} \in V$, $A[\mathbf{v}]_{\mathcal{B}} = [T(\mathbf{v})]_{\mathcal{C}}$, then $A = {}_{\mathcal{C}}[T]_{\mathcal{B}}$.

Proof of Fact 4.11. If $\mathbf{v} = x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n$, then $[\mathbf{v}]_{\mathcal{B}}$ is the column vector $\mathbf{x}$ with entries $x_1, x_2, \ldots, x_n$. Then $T(\mathbf{v}) = x_1T(\mathbf{v}_1) + x_2T(\mathbf{v}_2) + \cdots + x_nT(\mathbf{v}_n)$, and hence

$$\begin{aligned}[T(v)]_{\mathcal{C}} &= [x_1T(\mathbf{v}_1) + x_2T(\mathbf{v}_2) + \cdots + x_nT(\mathbf{v}_n)]_{\mathcal{C}} \\ &= x_1[T(\mathbf{v}_1)]_{\mathcal{C}} + x_2[T(\mathbf{v}_2)]_{\mathcal{C}} + \cdots + x_n[T(\mathbf{v}_n)]_{\mathcal{C}}\end{aligned}$$

(where it becomes apparent again that the coordinate mapping is a linear transformation). But this last expression is precisely the right hand side of the equation we are trying to prove. Thus, we are done. □

This is best understood by considering the coordinate mapping: we have a one-to-one correspondence, the so called, coordinate map, $V \to \mathbb{F}^n$, which associated to each $\mathbf{v}$ its coordinate vector $[\mathbf{v}]_{\mathcal{B}}$.

Similarly, we have a one-to-one correspondence $W \to \mathbb{F}^m$, which maps $\mathbf{w}$ to $[\mathbf{w}]_{\mathbb{C}}$. Both these mappings are linear transformations (and they are also both bijective (one-one and onto)).

Both these correspondences are linear transformations. The linear transformation T can now be analyzed by the following diagram (where $A = {}_{\mathcal{C}}[T]_{\mathcal{B}}$):

$$\begin{tikzcd} V \arrow[r, "{[\ ]_{\mathcal{B}}}"] \arrow[d, "T"'] & \mathbb{F}^n \arrow[d, "T_A"] \\ W \arrow[r, "{[\ ]_{\mathcal{C}}}"'] & \mathbb{F}^m \end{tikzcd}$$

Since coordinate mappings are one-to-one correspondences, T_A and T "look" very much alike. For instance, T is injective iff T_A is. The null spaces of A and T correspond to each other in this diagram. So to compute $T(\mathbf{v})$, we could first compute $[\mathbf{v}]_{\mathcal{B}}$, then multiply by A, to get $\mathbf{y} = [T(\mathbf{v})]_{\mathcal{C}} = A[\mathbf{v}]_{\mathcal{B}}$, and then simply compute the vector in W that corresponds to $\mathbf{y}$ under the coordinate mapping.

4.12 Example. Let $V = M_{2\times 2}(\mathbb{R})$, and $T\colon V \to V$ be defined as $T(A) = A^t$. Then T is a linear transformation. Indeed, $T(A + B) = (A + B)^t = A^t + B^t = T(A) + T(B)$ and $T(cA) = (cA)^t = cA^t = cT(A)$.

Let $\mathcal{B} = (\mathbf{e}_{11}, \mathbf{e}_{12}, \mathbf{e}_{21}, \mathbf{e}_{22})$ be the standard basis of V. Note that $T(\mathbf{e}_{11}) = e_{11}$ and $T(\mathbf{e}_{22}) = \mathbf{e}_{22}$, whereas $T(\mathbf{e}_{12}) = e_{21}$ and $T(\mathbf{e}_{21}) = \mathbf{e}_{12}$. Hence,

$${}_{\mathcal{B}}[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

So for instance if

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

then

$$[T(A)]_{\mathcal{B}} = {}_{\mathcal{B}}[T]_{\mathcal{B}} \begin{bmatrix}1\\2\\0\\1\end{bmatrix} = \begin{bmatrix}1\\0\\2\\1\end{bmatrix} = [A^t]_{\mathcal{B}}.$$

If $\mathcal{B}$ is a basis for an $\mathbb{F}$-vector space V and if W is another $\mathbb{F}$-vector space, then *any* function $f\colon \mathcal{B} \to W$ can be *extended* to a linear transformation $T\colon V \to W$ in a unique way: if $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$, then

$$T(x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_n\mathbf{v}_n) = x_1 f(\mathbf{v}_1) + x_2 f(\mathbf{v}_2) + \cdots + x_n f(\mathbf{v}_n).$$

(Can you verify that this is indeed a linear transformation, and the only one that extends f?) In particular, if $\mathcal{C} = (\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m)$ is a basis for W, we can realize any $m \times n$-matrix A with entries in $\mathbb{F}$ as the matrix of a unique linear transformation $T\colon V \to W$:

$$T(\mathbf{v}) := \mathcal{C}A[\mathbf{v}]_{\mathcal{B}}$$

does the trick. (This fits into the above scheme with $f\colon \mathcal{B} \to W$ defined as $f(\mathbf{v}_i) = \mathcal{C}\mathbf{a}_i$, where $\mathbf{a}_i$ is the i-th column of A.)

4.13 Example. Let V be a real vector space with basis $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4)$ and let W be a vector space with basis $(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3)$.

For *any* 3×4-matrix A, there is a *unique* linear transformation $T\colon V \to W$ such that $A = {}_{\mathcal{C}}[T]_{\mathcal{B}}$.

As an example consider

$$A = \begin{bmatrix}1 & 1 & 1 & 1\\0 & 2 & 2 & 1\\1 & 2 & -1 & 1\end{bmatrix}$$

Then define

$$\begin{aligned}T(\mathbf{v}_1) &= \mathbf{w}_1 + \mathbf{w}_3\\T(\mathbf{v}_2) &= \mathbf{w}_1 + 2\mathbf{w}_2 + 2\mathbf{w}_3\\T(\mathbf{v}_3) &= \mathbf{w}_1 + 2\mathbf{w}_2 - \mathbf{w}_3\\T(\mathbf{v}_4) &= \mathbf{w}_1 + \mathbf{w}_2 + \mathbf{w}_3.\end{aligned}$$

We have specified T on the basis $\mathcal{B}$. For an arbitrary $\mathbf{v} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + x_3\mathbf{v}_3 + x_4\mathbf{v}_4$, we then define

$$T(\mathbf{v}) = x_1T(\mathbf{v}_1) + x_2T(\mathbf{v}_2) + x_3T(\mathbf{v}_3) + x_4T(\mathbf{v}_4).$$

It is now elementary to verify that T is indeed a linear transformation and that $A = {}_{\mathcal{C}}[T]_{\mathcal{B}}$.

For instance, if $\mathbf{v} = \mathbf{v}_1 + 2\mathbf{v}_2 + \mathbf{v}_4$, then

$$[T(\mathbf{v})]_{\mathcal{C}} = A\begin{bmatrix}1\\2\\0\\1\end{bmatrix} = \begin{bmatrix}4\\5\\6\end{bmatrix}$$

so $T(\mathbf{v}) = 4\mathbf{w}_1 + 5\mathbf{w}_2 + 6\mathbf{w}_3$.

What is the null space of T? Well, $\mathbf{v} \in \mathrm{Nul}(T)$ iff $A[\mathbf{v}]_{\mathcal{B}} = \mathbf{0}$, so $\mathrm{Nul}(T)$ corresponds to $\mathrm{Nul}(A)$ via the coordinate mapping. A basis for $\mathrm{Nul}(A)$ thus gives a basis for T as we have seen many times now.

What is the image of T? $\mathbf{w} \in W$ is an element of $\mathrm{im}(T)$, iff $[\mathbf{w}]_{\mathcal{C}}$ is an element of $\mathrm{Col}(A)$.

X Problem. Let $T\colon U \to V$ and $S\colon V \to W$ be linear transformation between $\mathbb{F}$-vector spaces. Let $\mathcal{B}$ be a basis for U, $\mathcal{C}$ one for V and $\mathcal{D}$ one for W. Show that

$$_{\mathcal{D}}[S \circ T]_{\mathcal{B}} = {_{\mathcal{D}}[S]_{\mathcal{C}}} \cdot {_{\mathcal{C}}[T]_{\mathcal{B}}}.$$

(*Hint*: Verify that for all $\mathbf{u} \in U$, both sides map $[\mathbf{u}]_{\mathcal{B}}$ to $[S \circ T(\mathbf{u})]_{\mathcal{D}}$.)

4.2.2. Change of basis

Suppose we want to change the basis in a vector space V with basis $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and change it to a basis $\mathcal{C} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$.

Of course, if $\mathbf{v} \in V$, generally $[\mathbf{v}]_{\mathcal{B}} \neq [\mathbf{v}]_{\mathcal{C}}$. How can we determine $[\mathbf{v}]_{\mathcal{C}}$? Consider the linear transformation $T = \mathrm{id}\colon V \to V$. It is defined by $T(\mathbf{v}) = \mathbf{v}$. By the previous section, the matrix of T with respect to the bases $\mathcal{B}$ and $\mathcal{C}$ satisfies:

$$[\mathbf{v}]_{\mathcal{C}} = [T(\mathbf{v})]_{\mathcal{C}} = {_{\mathcal{C}}[T]_{\mathcal{B}}}[\mathbf{v}]_{\mathcal{B}}$$

So let $_{\mathcal{C}}P_{\mathcal{B}}$ be the matrix of T with respect to the bases $\mathcal{B}$ and $\mathcal{C}$. Then its ith column is given by $[\mathbf{v}_i]_{\mathcal{C}}$. Thus $_{\mathcal{C}}P_{\mathcal{B}}$ expresses the *old* basis $\mathcal{B}$ in terms of the *new* basis $\mathcal{C}$.

$$_{\mathcal{C}}P_{\mathcal{B}} = \begin{bmatrix} [\mathbf{v}_1]_{\mathcal{C}} & [\mathbf{v}_2]_{\mathcal{C}} & \dots & [\mathbf{v}_n]_{\mathcal{C}} \end{bmatrix}.$$

Note that usually the elements $\mathbf{w}_i$ of $\mathcal{C}$ will be specified in terms of their coordinate vectors with respect to $\mathcal{B}$, so we are usually *not* given $_{\mathcal{C}}P_{\mathcal{B}}$ but rather $_{\mathcal{B}}P_{\mathcal{C}}$. See also Example 4.8

4.14 Definition. The $n \times n$-matrix $_{\mathcal{C}}P_{\mathcal{B}}$ defined above is called the *change of basis matrix* corresponding to the change form $\mathcal{B}$ to $\mathcal{C}$.

We have

$$_{\mathcal{C}}P_{\mathcal{B}}[\mathbf{v}]_{\mathcal{B}} = [\mathbf{v}]_{\mathcal{C}}.$$

Thus, as soon as we know how to express the elements $\mathbf{v}_i$ in terms of the new coordinates, we can compute the new coordinates for any element.

Indeed, if $\mathbf{v} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_n\mathbf{v}_n$, then

$$[\mathbf{v}]_{\mathcal{C}} = x_1[\mathbf{v}_1]_{\mathcal{C}} + x_2[\mathbf{v}_2]_{\mathcal{C}} + \dots + x_n[\mathbf{v}_n]_{\mathcal{C}} = {_{\mathcal{C}}P_{\mathcal{B}}}[\mathbf{v}]_{\mathcal{B}}.$$

Notice that this immediately implies that

$$_{\mathcal{C}}P_{\mathcal{B}}[\mathbf{w}_i]_{\mathcal{B}} = [\mathbf{w}_i]_{\mathcal{C}} = \mathbf{e}_i.$$

Hence ${}_{\mathcal{C}}P_{\mathcal{B}}$ is invertible: if $Q = [\,[\mathbf{w}_1]_{\mathcal{B}}\,[\mathbf{w}_2]_{\mathcal{B}}\,\ldots\,[\mathbf{w}_n]_{\mathcal{B}}\,]$ is the matrix with columns $[\mathbf{w}_i]_{\mathcal{B}}$, then ${}_{\mathcal{C}}P_{\mathcal{B}}Q = I_n$.

How can we compute ${}_{\mathcal{C}}P_{\mathcal{B}}$? In practice we are often given the $[\mathbf{w}_i]_{\mathcal{B}}$ and hence Q. Also note that $Q = {}_{\mathcal{B}}P_{\mathcal{C}}$ is the change of basis matrix for the reverse change.

4.15 Fact. ${}_{\mathcal{B}}P_{\mathcal{C}} = {}_{\mathcal{C}}P_{\mathcal{B}}^{-1}$.

4.16 Example. Let $V = \mathbb{R}^2$ and let $\mathcal{B} = (\mathbf{e}_1, \mathbf{e}_2)$ be the standard basis. Let

$$\mathcal{C} = \left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right)$$

Compute the change of basis matrix for the change form $\mathcal{B}$ to $\mathcal{C}$. Compute $[\mathbf{x}]_{\mathcal{C}}$ where

$$\mathbf{x} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

Notice that since $[\mathbf{v}]_{\mathcal{B}} = \mathbf{v}$ we are given ${}_{\mathcal{B}}P_{\mathcal{C}}$ for free:

$${}_{\mathcal{B}}P_{\mathcal{C}} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

By the above

$${}_{\mathcal{C}}P_{\mathcal{B}} = ({}_{\mathcal{B}}P_{\mathcal{C}})^{-1} = \begin{bmatrix} \frac{-1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{-1}{3} \end{bmatrix}.$$

We get

$$[\mathbf{x}]_{\mathcal{C}} = \begin{bmatrix} \frac{-1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{-1}{3} \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{5}{3} \\ \frac{-1}{3} \end{bmatrix}.$$

Let us check:

$$\frac{5}{3} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \frac{-1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \mathbf{x},$$

so our computation is consistent.

Finally, if e.g.

$$\mathcal{C}' = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)$$

find ${}_{\mathcal{C}'}P_{\mathcal{C}}$. Note that

$${}_{\mathcal{C}'}P_{\mathcal{C}} = {}_{\mathcal{C}'}P_{\mathcal{B}\,\mathcal{B}}P_{\mathcal{C}}.$$

Indeed,

$${}_{\mathcal{C}'}P_{\mathcal{B}\,\mathcal{B}}P_{\mathcal{C}}[\mathbf{v}]_{\mathcal{C}} = {}_{\mathcal{C}'}P_{\mathcal{B}}[\mathbf{v}]_{\mathcal{B}} = [\mathbf{v}]_{\mathcal{C}'}.$$

This is true for all $[\mathbf{v}]_{\mathcal{C}'}$ and hence also for the elements of $\mathcal{B}$ and so ${}_{\mathcal{C}'}P_{\mathcal{C}} = {}_{\mathcal{C}'}P_{\mathcal{B}\,\mathcal{B}}P_{\mathcal{C}}$. Here

$${}_{\mathcal{C}'}P_{\mathcal{B}} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

and so

$$_{\mathcal{C}'}P_{\mathcal{C}} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$$

4.17 Example. Let $V = \mathcal{P}(\mathbb{R})_3$. Let $\mathcal{B} = (1, x, x^2, x^3)$ be its standard basis.

Let f_0, f_1, f_2, f_3 be defined as

$$f_i = \prod_{\substack{j=0 \\ j \neq i}}^{3} \frac{x-j}{i-j}.$$

So

$$\begin{aligned} f_0 &= \frac{x-1}{0-1}\frac{x-2}{0-2}\frac{x-3}{0-3} \\ f_1 &= \frac{x}{1-0}\frac{x-2}{1-2}\frac{x-3}{1-3} \\ f_2 &= \frac{x}{2-0}\frac{x-1}{2-1}\frac{x-3}{2-3} \\ f_3 &= \frac{x}{3-0}\frac{x-1}{3-1}\frac{x-2}{3-2} \end{aligned}$$

Notice that for $j = 0, 1, 2, 3$:

$$f_i(j) = \begin{cases} 1 & j = i \\ 0 & j \neq i \end{cases} \tag{4.2}$$

$\mathcal{C} = (f_0, f_1, f_2, f_3)$ is linearly independent: indeed, suppose

$$c_0 f_0 + c_1 f_1 + c_2 f_2 + c_3 f_3 = \mathbf{0}.$$

Then this is true when evaluated at each real number. So we may consider for instance $x = 0$:

$$c_0 f(0) + c_1 f_1(0) + c_2 f_2(0) + c_3 f_3(0) = 0.$$

By (4.2) the left hand side is equal to $c_0 \cdot 1$, so $c_0 = 0$. Similarly, evaluating at $x = i$ gives us $c_i = 0$ for $i = 1, 2, 3$. Since $\dim V = 4$ this means $\mathcal{C}$ is a basis.

Let us compute $_{\mathcal{C}}P_{\mathcal{B}}$.

Notice that for any $f \in V$ we have

$$f = f(0)f_0 + f(1)f_1 + f(2)f_2 + f(3)f_3.$$

Indeed, $\mathcal{C}$ is a basis so any f can be written as $f = a_0 f_0 + a_1 f_1 + a_2 f_2 + a_3 f_3$, so $f(i) = a_i f(i) = a_i$, using (4.2) once more.

Thus,

$$_{\mathcal{C}}P_{\mathcal{B}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 2^2 & 2^3 \\ 1 & 3 & 3^2 & 3^3 \end{bmatrix}$$

and for $f = c_0 + c_1x + c_2x^2 + c_3x^3$ we have

$$[f]_{\mathcal{C}} = \begin{bmatrix} f(0) \\ f(1) \\ f(2) \\ f(3) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1c_0 \\ c_0 + c_1 + c_2 + c_3 \\ 1c_0 + 2c_1 + 4c_2 + 8c_3 \\ 1c_0 + 3c_1 + 9c_3 + 27c_3 \end{bmatrix} = [f]_{\mathcal{C}}$$

As an application, this allows us to find a (unique) polynomial of degree at most 3 with prescribed values at positions $0, 1, 2, 3$. We get its coordinate vector $[f]_{\mathcal{C}}$ for free (it is given by exactly the prescribed values). But usally we want a polynomial to be given in terms of the standard basis, so for this, we could compute $[f]_{\mathcal{B}} = {}_{\mathcal{C}}P_{\mathcal{B}}^{-1}[f]_{\mathcal{C}}$.

Finally, how does ${}_{\mathcal{C}}[T]_{\mathcal{B}}$ change? Suppose we change the basis in W form $\mathcal{C}$ to $\mathcal{C}'$ and the basis of V to $\mathcal{B}' = (\mathbf{v}'_1, \mathbf{v}'_2, \ldots, \mathbf{v}'_n)$ from $\mathcal{B}$. For any $\mathbf{v}$ in V, we have:

$$[T(\mathbf{v})]_{\mathcal{C}'} = {}_{\mathcal{C}'}P_{\mathcal{C}}[T(\mathbf{v})]_{\mathcal{C}} = {}_{\mathcal{C}'}P_{\mathcal{C}\mathcal{C}}[T]_{\mathcal{B}}[\mathbf{v}]_{\mathcal{B}} = {}_{\mathcal{C}'}P_{\mathcal{C}\mathcal{C}}[T]_{\mathcal{B}}({}_{\mathcal{B}'}P_{\mathcal{B}})^{-1}[\mathbf{v}]_{\mathcal{B}'}.$$

(Note we could have used ${}_{\mathcal{B}}P_{\mathcal{B}'}$ above to avoid the inverse; however, we want to formulate everything in terms of the change of basis matrix, and this is ${}_{\mathcal{B}'}P_{\mathcal{B}}$ if $\mathcal{B}$ is the old, and $\mathcal{B}'$ is the new basis.)

Since this is true for all $\mathbf{v}$ and hence for all $\mathbf{v}'_i \in \mathcal{B}'$, we conclude that

$${}_{\mathcal{C}'}[T]_{\mathcal{B}'} = {}_{\mathcal{C}'}P_{\mathcal{C}\mathcal{C}}[T]_{\mathcal{B}}({}_{\mathcal{B}'}P_{\mathcal{B}})^{-1}$$

To get a more readable form, let us write $\mathbf{x}$ for the old coordinates of a given vector $\mathbf{v}$, $\mathbf{x}'$ for the new ones, so that $\mathbf{x} = [\mathbf{v}]_{\mathcal{B}}$, $\mathbf{x}' = [\mathbf{v}]_{\mathcal{B}'}$. Similarly, let $\mathbf{y} = [T(\mathbf{v})]_{\mathcal{C}}$ and $\mathbf{y}' = [T(\mathbf{v})]_{\mathcal{C}'}$, and finally put $A = {}_{\mathcal{C}}[T]_{\mathcal{B}}$ and $B = {}_{\mathcal{C}'}[T]_{\mathcal{B}'}$. Then

$$\begin{aligned} \mathbf{y} &= A\mathbf{x} \\ \mathbf{y}' &= B\mathbf{x}' \end{aligned}$$

with

$$B = QAP^{-1}$$

where

$$\begin{aligned} Q &= {}_{\mathcal{C}'}P_{\mathcal{C}} \\ P &= {}_{\mathcal{B}'}P_{\mathcal{B}} \end{aligned}$$

are the change of basis matrices for the changes from the "old" to the "new" bases.

What about a case where we have a vector space V with *three* bases $\mathcal{B}, \mathcal{C}, \mathcal{D}$? Are the change of basis matrices related?

4.18 Fact. Let V be a vector space with (finite) bases $\mathcal{B}, \mathcal{C}, \mathcal{D}$. Then

$${}_{\mathcal{D}}P_{\mathcal{C}\mathcal{C}}P_{\mathcal{B}} = {}_{\mathcal{D}}P_{\mathcal{B}}.$$

Proof. This is a consequence of the result about compositions of transformations (cf. Problem X), but we can repeat the argument.

${}_{\mathcal{D}}P_{\mathcal{B}}$ is the unique matrix A such that

$$[\mathbf{v}]_{\mathcal{D}} = A[x]_{\mathcal{B}}$$

for all $\mathbf{v} \in V$. It therefore suffices to check that ${}_{\mathcal{D}}P_{\mathcal{C}\mathcal{C}}P_{\mathcal{B}}$ also has that property: let $\mathbf{v} \in V$, then

$${}_{\mathcal{D}}P_{\mathcal{C}\mathcal{C}}P_{\mathcal{B}}[v]_{\mathcal{B}} = {}_{\mathcal{D}}P_{\mathcal{C}}[\mathbf{v}]_{\mathcal{C}} = [\mathbf{v}]_{\mathcal{D}}$$

as needed. □

4.19 Example. Suppose we want to find ${}_{\mathcal{D}}P_{\mathcal{C}}$ but are given ${}_{\mathcal{B}}P_{\mathcal{C}}$ and ${}_{\mathcal{B}}P_{\mathcal{D}}$. Then

$${}_{\mathcal{D}}P_{\mathcal{C}} = {}_{\mathcal{D}}P_{\mathcal{B}\mathcal{B}}P_{\mathcal{C}} = ({}_{\mathcal{B}}P_{\mathcal{D}})^{-1}{}_{\mathcal{B}}P_{\mathcal{C}}.$$

Consider a vector space V with basis $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2)$, and let

$$\begin{aligned}
\mathbf{w}_1 &= \mathbf{v}_1 + \mathbf{v}_2 \\
\mathbf{w}_2 &= \mathbf{v}_1 + 2\mathbf{v}_2 \\
\mathbf{u}_1 &= 2\mathbf{v}_1 - \mathbf{v}_2 \\
\mathbf{u}_2 &= 2\mathbf{v}_1 + \mathbf{v}_2
\end{aligned}$$

Then $\mathcal{C} := (\mathbf{w}_1, \mathbf{w}_2)$ and $\mathcal{D} := (\mathbf{u}_1, \mathbf{u}_2)$ are bases for V. We can also read off

$$P := {}_{\mathcal{B}}P_{\mathcal{C}} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \text{ and } Q := {}_{\mathcal{B}}P_{\mathcal{D}} = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$$

We conclude that

$${}_{\mathcal{D}}P_{\mathcal{C}} = Q^{-1}P = \frac{1}{4}\begin{bmatrix} 1 & -2 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} -1 & -3 \\ 3 & 5 \end{bmatrix}$$

Note this says that

$$\begin{aligned}
\mathbf{w}_1 &= \frac{-1}{4}\mathbf{u}_1 + \frac{3}{4}\mathbf{u}_2 \\
\mathbf{w}_2 &= \frac{-3}{4}\mathbf{u}_1 + \frac{5}{4}\mathbf{u}_2
\end{aligned}$$

which we can verify by inspection.

4.3. Rank and nullity for linear transformations

4.20 Theorem (Rank-Nullity Theorem for Linear Transformations)**.** *Let $T\colon V \to W$ be a linear transformation and suppose $\dim V = n$. Then*

$$\dim \operatorname{Nul}(T) + \dim \operatorname{im}(T) = n.$$

4.21 Definition. The *rank* $\operatorname{rank}(T)$ of a linear transformation T is the dimension of $\operatorname{im}(T)$. The *nullity* $\operatorname{nullity}(T)$ of T is the dimension of $\operatorname{Nul}(T)$.

The theorem therefore states that

$$\operatorname{rank}(T) + \operatorname{nullity}(T) = \dim V.$$

Note it should be clear that if V, W are both finite dimensional and $\mathcal{B}$ is a basis for V and $\mathcal{C}$ is a basis for T then

$$\operatorname{rank}(T) = \operatorname{rank}{}_{\mathcal{C}}[T]_{\mathcal{B}}$$
$$\operatorname{nullity}(T) = \operatorname{nullity}{}_{\mathcal{C}}[T]_{\mathcal{B}}$$

The Rank-Nullity Theorem is then an immediate consequence (if $\dim W < \infty$). However, we will give an alternate proof that provides a stronger result.

4.22 Theorem. *Let $T\colon V \to W$ be a linear transformation where $\dim V = n$ and $\dim W = m$. Suppose $\operatorname{rank}(T) = r$, then there are bases $\mathcal{B}$ for V and $\mathcal{C}$ for W such that $A := {}_{\mathcal{C}}[T]_{\mathcal{B}}$ has the form*

$$A = \begin{bmatrix} I_r & 0_{r\times(n-r)} \\ 0_{(m-r)\times r} & 0_{(m-r)\times(n-r)} \end{bmatrix} \tag{4.3}$$

where I_r is the $r \times r$-identity matrix, and $0_{p\times q}$ is the $p \times q$ zero matrix with the understanding that if $r, p,$ or $q = 0$, then the corresponding matrices are just left off.

Proof. If $r = 0$, then $T(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v}$ and $A = 0_{m\times n}$. We may therefore assume that $r > 0$. Let $(\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_r)$ be a basis for $\operatorname{im}(T)$, and extend it to a basis $\mathcal{C} := (\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_r, \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{m-r})$ of W.[2]

Next, let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r \in V$ be elements such that $T(\mathbf{v}_i) = \mathbf{w}_i$. Such elements exist, because $\mathbf{w}_i \in \operatorname{im}(T)$ by definition. Let $(\mathbf{n}_1, \mathbf{n}_2, \ldots, \mathbf{n}_q)$ be a basis for $\operatorname{Nul}(T)$. Let $\mathcal{B} := (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r, \mathbf{n}_1, \mathbf{n}_2, \ldots, \mathbf{n}_p)$. Then $\mathcal{B}$ is a basis for V. To see this, let

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_r\mathbf{v}_r + d_1\mathbf{n}_1 + d_2\mathbf{n}_2 + \cdots + d_p\mathbf{n}_p = \mathbf{0}.$$

Applying T to both sides and using the fact that $T(\mathbf{n}_i) = 0$, we get

$$c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \cdots + c_rT(\mathbf{v}_r) = c_1\mathbf{w}_1 + \cdots + c_r\mathbf{w}_r = \mathbf{0}.$$

But $(\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_r)$ is linearly independent, and therefore $c_1 = c_2 = \cdots = c_r = 0$. This means that

$$d_1\mathbf{n}_1 + d_2\mathbf{n}_2 + \cdots + \mathbf{n}_p = \mathbf{0}.$$

Linear independence of the $\mathbf{n}_i$ then forces $d_1 = d_2 = \cdots = d_p = 0$ and $\mathcal{B}$ is linearly independent.

[2] If $\mathcal{D}$ is *any* basis of V, we could already conclude that the last $m - r$ rows of ${}_{\mathcal{C}}[T]_{\mathcal{D}}$ are zero. Why?

To see that $\mathcal{B}$ spans V, we observe that if $\mathbf{v} \in V$ then $T(v) = c_1\mathbf{w}_1 + c_2\mathbf{w}_2 + \cdots + c_r\mathbf{w}_r$ for some $c_1, c_2, \ldots, c_r$ and hence

$$T(\mathbf{v}) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \cdots + c_rT(\mathbf{v}_r) = T(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_r\mathbf{v}_r).$$

This means

$$\mathbf{v} - c_1\mathbf{v}_1 - c_2\mathbf{v}_2 - \cdots - c_r\mathbf{v}_r \in \mathrm{Nul}(T),$$

and hence

$$\mathbf{v} - c_1\mathbf{v}_1 - c_2\mathbf{v}_2 - \cdots - c_r\mathbf{v}_r = d_1\mathbf{n}_1 + d_2\mathbf{n}_2 + \cdots + d_p\mathbf{n}_p$$

for suitable $d_1, d_2, \ldots, d_p \in \mathbb{F}$. In other words

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_r\mathbf{v}_r + d_1\mathbf{n}_1 + d_2\mathbf{n}_2 + \cdots + d_p\mathbf{n}_p \in \mathrm{Span}(\mathcal{B}).$$

It remains to show that $A := {}_{\mathcal{C}}[T]_{\mathcal{B}}$ has the desired form. But this is a consequence of the fact that

$$[T(\mathbf{v}_i)]_{\mathcal{C}} = \mathbf{e}_i$$

for $i = 1, 2, \ldots, r$ where $\mathbf{e}_i$ is the i-th column of the $m \times m$ identity matrix, and

$$[T(\mathbf{n}_i)]_{\mathcal{C}} = \mathbf{0}$$

for $i = 1, 2, \ldots, p$. □

4.23 Corollary (of proof)**.** *We have $p + r = n$.*

4.24 Corollary (Matrix Version)**.** *Let A be an $m \times n$ matrix or rank r, then there is an invertible $m \times m$ matrix P and an invertible $n \times n$ matrix Q such that PAQ^{-1} as the form (4.3).*

Proof. Apply the theorem to the matrix transformation $T_A \colon \mathbb{F}^n \to \mathbb{F}^m$. Let $\mathcal{E}_n$ and $\mathcal{E}_m$ be the standard bases of $\mathbb{F}^n$ and $\mathbb{F}^m$ respectively, then in the notation of the theorem, we may pick $P = {}_{\mathcal{C}}P_{\mathcal{E}_m}$ and $Q = {}_{\mathcal{B}}P_{\mathcal{E}_n}$. Then

$${}_{\mathcal{C}}[T_A]_{\mathcal{B}} = PAQ^{-1}$$

has the desired form. □

4.25 Example. Let

$$A = \begin{bmatrix} 3 & -1 & 4 & 1 & 2 & 1 \\ 5 & 1 & 4 & 1 & 1 & 2 \\ 7 & 2 & 5 & 1 & 1 & 3 \\ 9 & 3 & 6 & 1 & 1 & 4 \\ 11 & 4 & 7 & 1 & 1 & 5 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ & 1 & -1 & 0 & -1 & 0 \\ & & & 1 & -\frac{1}{2} & -\frac{1}{2} \\ & & & & & 0 \\ & & & & & 0 \end{bmatrix}$$

If we label the columns of A by $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_6$, we can read off the following information:

- $\operatorname{rank} A = 3$;

- $\operatorname{nullity} A = 3$;
- a basis for $\operatorname{Col}(A) = \operatorname{im}(T_A)$ is given by $(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_4)$;
- a basis for $\operatorname{Nul}(A) = \ker T_A$ is given by

$$\left(\mathbf{n}_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \mathbf{n}_2 = \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \\ \frac{1}{2} \\ 1 \\ 0 \end{bmatrix}, \mathbf{n}_3 = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \\ 0 \\ 1 \end{bmatrix} \right)$$

The (proof of the) theorem says that since $T(\mathbf{e}_1) = \mathbf{a}_1$, $T(\mathbf{e}_2) = \mathbf{a}_2$, and $T(\mathbf{e}_4) = \mathbf{a}_4$ that

$$\mathcal{B} := \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \\ \frac{1}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \\ 0 \\ 1 \end{bmatrix} \right)$$

is a basis for $\mathbb{F}^6$. To extend the basis for $\operatorname{Col}(A)$ to one for $\mathbb{F}^5$, we compute

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_4 & I_5 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} & 0 & 0 & -\frac{5}{2} & 2 \\ & 1 & 0 & -1 & 0 & 0 & 4 & -3 \\ & & 1 & -\frac{3}{2} & 0 & 0 & \frac{23}{2} & -9 \\ & & & & 1 & 0 & -3 & 2 \\ & & & & & 1 & -2 & 1 \end{bmatrix}$$

to conclude that $\mathcal{C} = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_4, \mathbf{e}_2, \mathbf{e}_3)$[3] is a basis for $\mathbb{F}^5$.

With respect to the bases $\mathcal{B}$ and $\mathcal{C}$, T_A has the matrix

$$PAQ^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

[3] Keep in mind that the meaning of $\mathbf{e}_i$ depends on which $\mathbb{F}^n$ it is a member of.

Here $P = {}_{\mathcal{C}}P_{\mathcal{E}_5} = \begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_4 & \mathbf{e}_2 & \mathbf{e}_3\end{bmatrix}^{-1}$, and

$$Q = {}_{\mathcal{B}}P_{\mathcal{E}_6} \text{ is the } \textbf{inverse} \text{ of } \begin{bmatrix}1 & 0 & 0 & -1 & -\frac{1}{2} & -\frac{1}{2}\\ 0 & 1 & 0 & 1 & 1 & 0\\ 0 & 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2}\\ 0 & 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

Recall that a function $f\colon X \to Y$ is *surjective* if for every $y \in Y$ there is $x \in X$ such that $f(x) = y$. Thus, a linear transformation $T\colon V \to W$ is surjective if and only if $\operatorname{im}(T) = W$, or, equivalently (for finite dimensional vector spaces) $\operatorname{rank}(T) = \dim W$.

4.26 Definition. A linear transformation $T\colon V \to W$ of $\mathbb{F}$-vector spaces is called an *isomorphism* if it is bijective. We call V isomorphic to W if there is an isomorphism $T\colon V \to W$.

4.27 Remarks.

1. $T\colon V \to W$ is an isomorphism if and only if $\operatorname{Nul}(T) = \{\mathbf{0}\}$ and $\operatorname{im}(T) = W$.
2. T is an isomorphism if and only if $T_A\colon \mathbb{F}^n \to \mathbb{F}^m$ is where $A = {}_{\mathcal{C}}[T]_{\mathcal{B}}$ and $\mathcal{B}$ is a basis for V and $\mathcal{C}$ is a basis for W.
3. If T is an isomorphism then $\operatorname{rank} T = \dim W$, and $\operatorname{nullity} T = 0$; by the Rank Nullity Theorem this means that there are bases $\mathcal{B}$ and $\mathcal{C}$ for V and W, reaspectively, such that ${}_{\mathcal{C}}[T]_{\mathcal{B}}$ is the $n \times n$ identity matrix where $n = \dim V = \dim W$.

4.28 Fact. Let $T\colon V \to W$ be an isomorphism of $\mathbb{F}$-vector spaces. Then the inverse function $T^{-1}\colon W \to V$ is also a linear transformation. Moreover, if $\mathcal{B}$ is a basis for V and $\mathcal{C}$ is a basis for W then

$${}_{\mathcal{B}}[T^{-1}]_{\mathcal{C}} = ({}_{\mathcal{C}}[T]_{\mathcal{B}})^{-1}.$$

Proof. Exercise. □

Because of fact being isomorphic is symmetric: if V is isomorphic to W then also W is isomorphic to V, and we simply say "V and W are isomorphic."

4.29 Theorem. *Let V, W be two (finite-dimensional)*[4] *$\mathbb{F}$-vector spaces, then V and W are isomorphic if and only if $\dim V = \dim W$.*

Proof. Suppose $\dim V = \dim W = n$, say. Let $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ and $\mathcal{C} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$ be bases for V and W respectively. Let $T\colon V \to W$ be the (unique) linear transformation such that $T(\mathbf{v}_i) = \mathbf{w}_i$ for $i = 1, 2, \dots, n$. (Recall that $T(\mathbf{v}) = \mathcal{C}[\mathbf{v}]_{\mathcal{B}}$.) Then T is an isomorphism: its inverse function is the linear transformation $S\colon W \to V$ that maps $\mathbf{w}_i$ to $\mathbf{v}_i$: $S(\mathbf{w}) = \mathcal{B}[\mathbf{w}]_{\mathcal{C}}$. Note one could also argue that T is injective ($\operatorname{Nul}(T) = \{\mathbf{0}\}$) and surjective ($\operatorname{im} T = W$).

Conversely if V, W are isomorphic we know that $\dim V = \dim W$ (cf. Remark 4.273). □

[4] The theorem holds in general.